Neural Network Accelerator

Team 11

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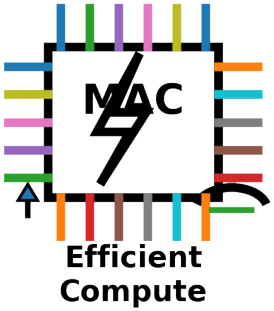
Importance & Design Goals

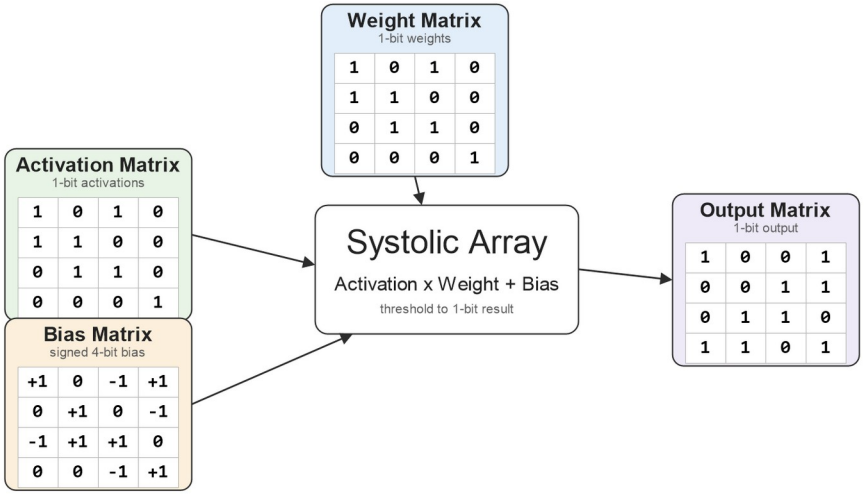
- Our chip is a small hardware accelerator for **matrix multiplication**, which is the core operation used in many neural network and AI workloads
- The chip takes **an activation matrix, a weight matrix, and a bias matrix,** then computes the output matrix using a compact binary systolic array and binary quantization.
- This type of accelerator can be used in low-power edge AI applications such as edge inference, image classification, sensor processing, and embedded neural network systems.

Matrix Multiplication

- Implement Matrix multiplication using a **systolic array architecture**, which uses parallelism for highly increased throughput
- Add an activation stage (ReLU) + quantization, then store results in an output memory with serial I/O interface support.

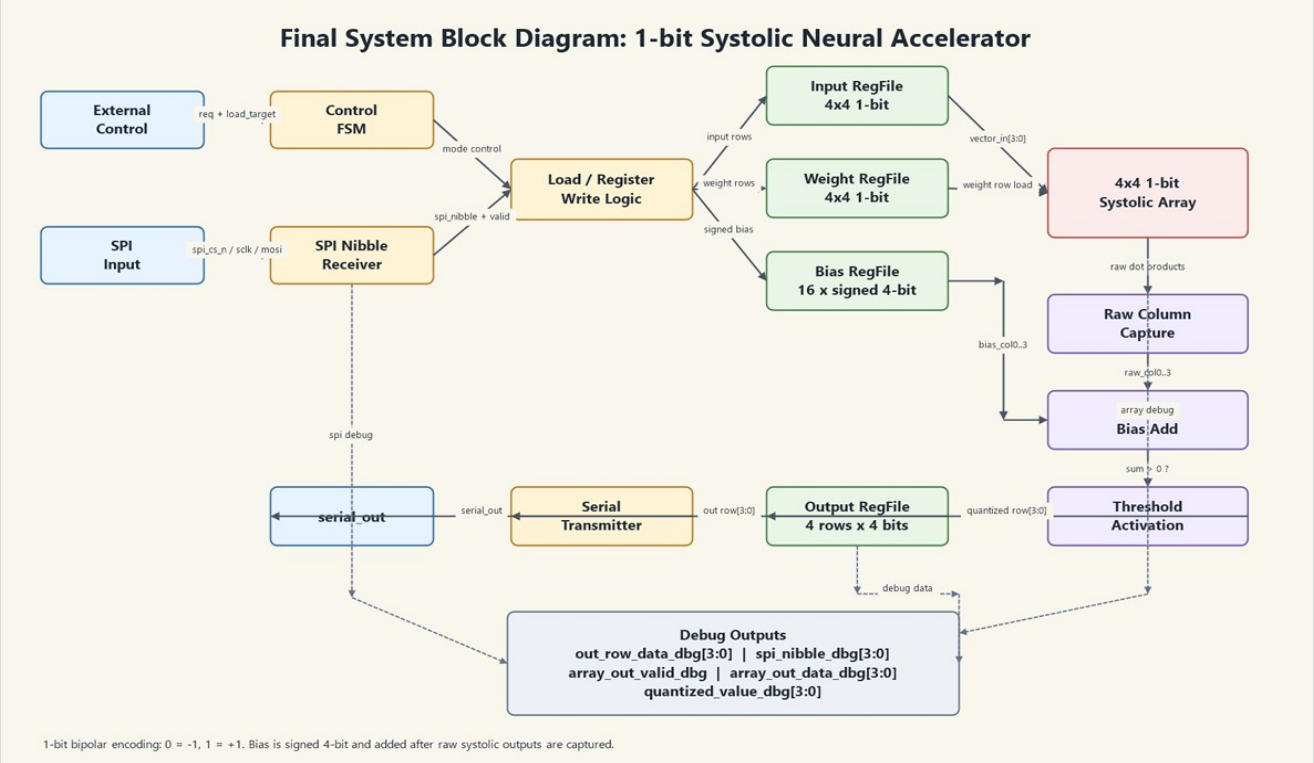






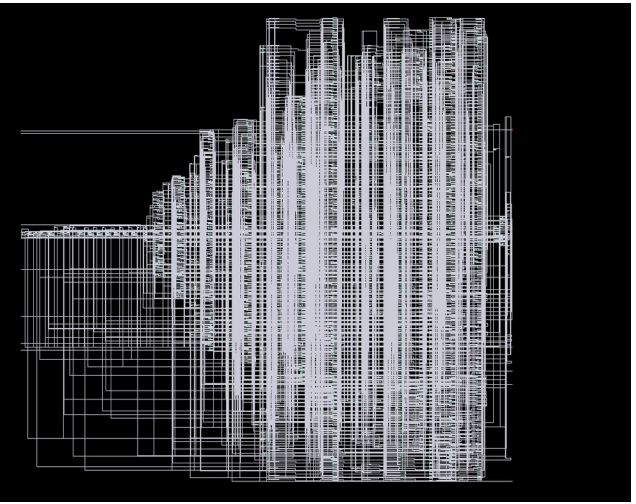
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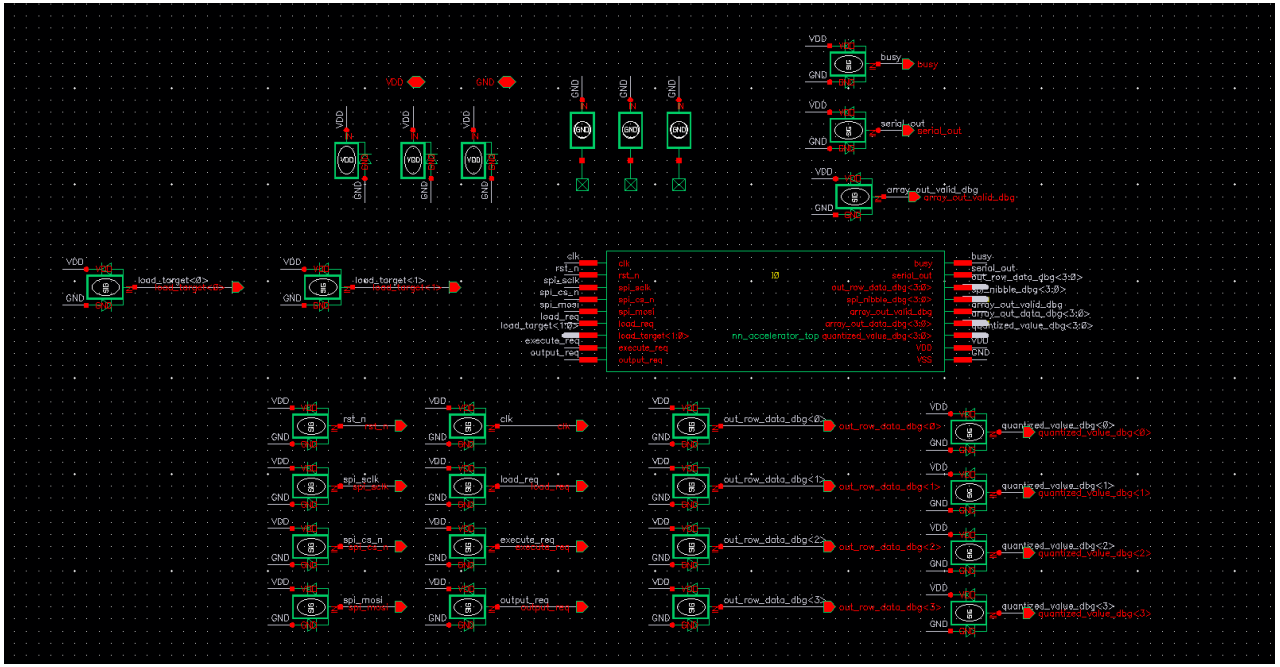
Block Diagram



Top Level Schematics

Import gate level .v netlist and .def physical layout files into virtuoso



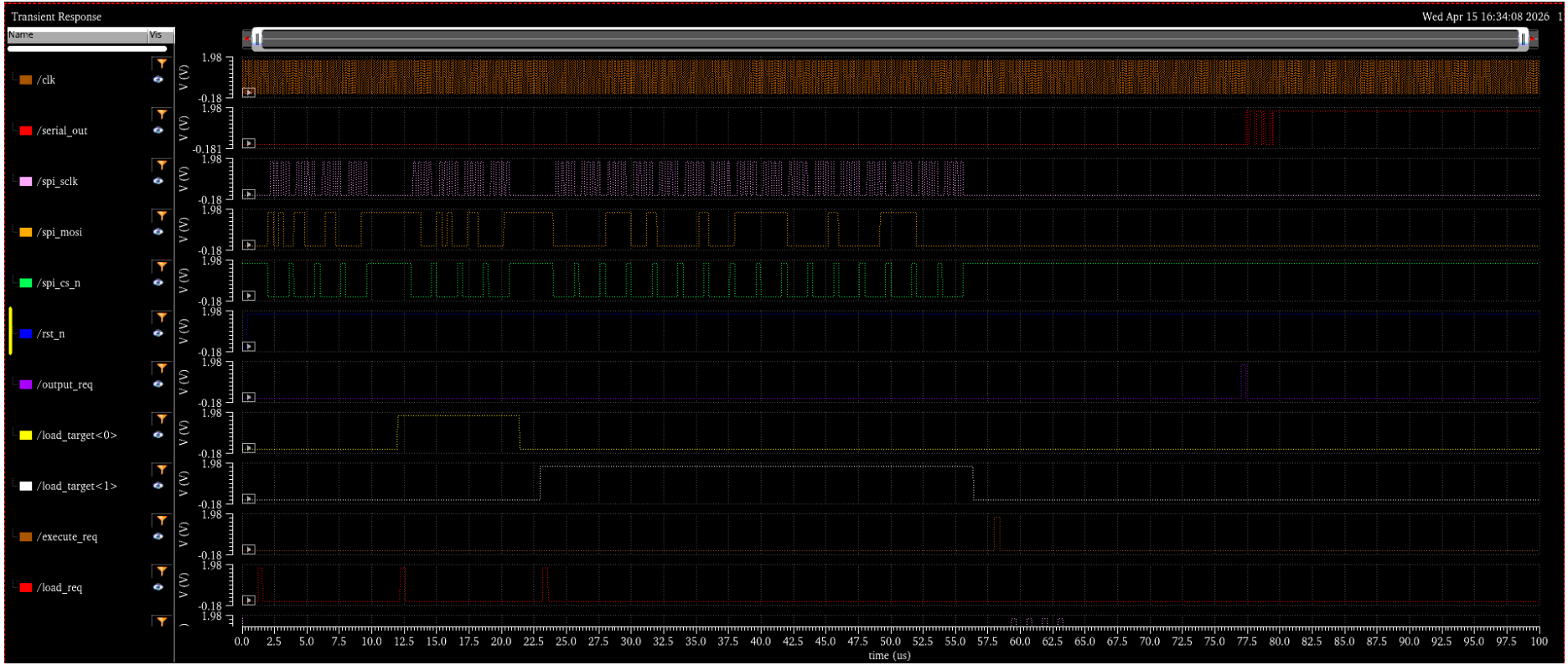


Schematic generated by Genus

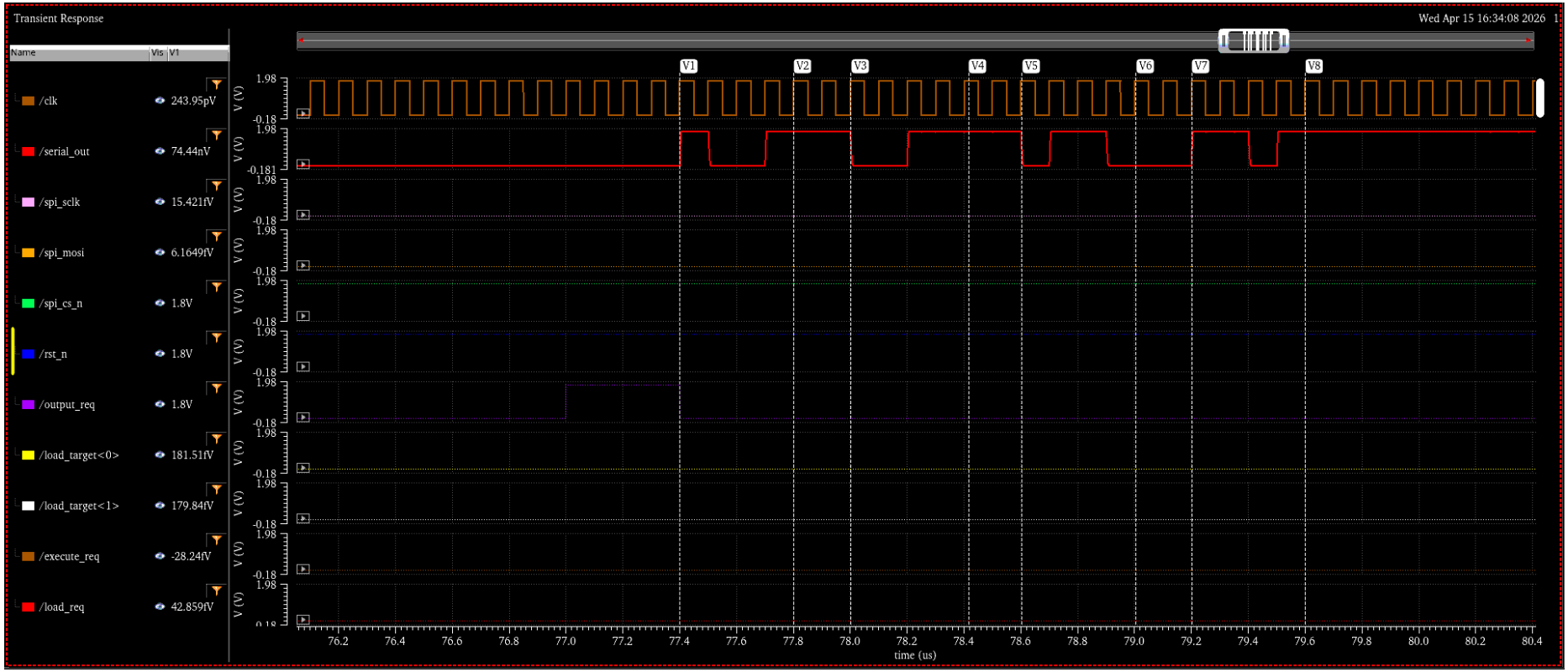
Top-level schematic

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Functional Simulation



Functional Simulation



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Power/Area Reports

```
Instance: /nn_accelerator_top
Power Unit: W
PDB Frames: /stim#0/frame#0
  ------------------------------------------------------------------------
    Category         Leakage     Internal    Switching        Total    Row%
  ------------------------------------------------------------------------
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
    register     6.12185e-08  2.94930e-04  2.72736e-06  2.97719e-04  88.65%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       logic     2.19420e-08  1.08493e-05  7.66011e-06  1.85313e-05   5.52%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       clock     0.00000e+00  0.00000e+00  1.95696e-05  1.95696e-05   5.83%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ------------------------------------------------------------------------
    Subtotal     8.31605e-08  3.05780e-04  2.99571e-05  3.35820e-04 100.00%
  Percentage           0.02%       91.05%        8.92%      100.00% 100.00%
  ------------------------------------------------------------------------
```

~.336 mW power consumption

```
-------------------------------------------------------
a211oi_1         16     999.040    gpdk_tt_1p8v_25c
a21oi_1          68    3232.720    gpdk_tt_1p8v_25c
a22oi_1         211   14037.830    gpdk_tt_1p8v_25c
a31oi_1           3     197.490    gpdk_tt_1p8v_25c
dfxtp           302  101314.963    gpdk_tt_1p8v_25c
inv_1           276    5017.680    gpdk_tt_1p8v_25c
nand2_1         166    5727.000    gpdk_tt_1p8v_25c
nand2b_1         18     905.220    gpdk_tt_1p8v_25c
nand3_1          26    1342.640    gpdk_tt_1p8v_25c
nand3b_1         14     930.300    gpdk_tt_1p8v_25c
nor2_1          283    9375.790    gpdk_tt_1p8v_25c
nor3_1           34    1696.600    gpdk_tt_1p8v_25c
o21a_1            6     382.200    gpdk_tt_1p8v_25c
o21ai_1         121    6275.060    gpdk_tt_1p8v_25c
xor2_1            7     676.620    gpdk_tt_1p8v_25c
-------------------------------------------------------
total          1551  152111.154


     Type      Instances    Area    Area %
-----------------------------------------
sequential           302 101314.963   66.6
inverter             276   5017.680    3.3
logic                973  45778.510   30.1
physical_cells         0      0.000    0.0
-----------------------------------------
total                1551 152111.154  100.0
```

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Chip finalization

```
module cmd_controller_4x4 (
    input  wire        clk,
    input  wire        rst_n,

    input  wire [7:0]  rx_byte,
    input  wire        rx_valid,

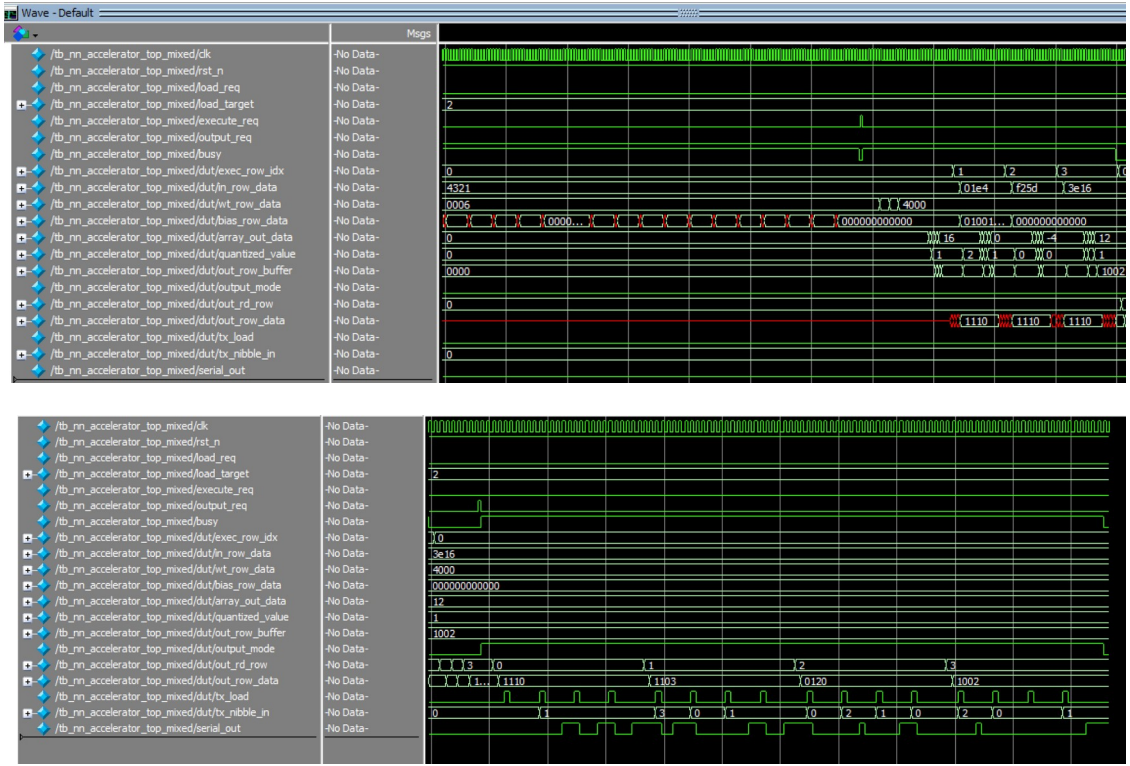
    // writes to activation SRAM
    output reg         act_wr_pulse,
    output reg  [1:0]  act_wr_row,
    output reg  [1:0]  act_wr_col,
    output reg  [7:0]  act_wr_data,

    // writes to weight SRAM
    output reg         wgt_wr_pulse,
    output reg  [1:0]  wgt_wr_row,
    output reg  [1:0]  wgt_wr_col,
    output reg  [7:0]  wgt_wr_data,

    // writes to bias (24-bit)
    output reg         bias_wr_pulse,
    output reg  [1:0]  bias_col,
    output reg  [23:0] bias_data,

    // control pulses
    output reg         start_compute_pulse,
    output reg         start_output_pulse
```

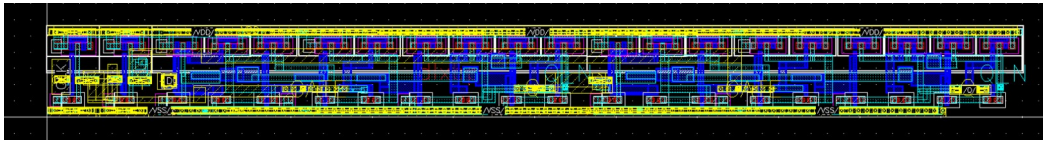
Verilog code



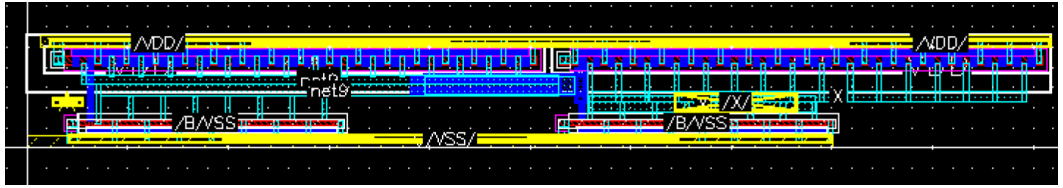
Top level RTL simulation

Chip finalization

- Characterize standard cell library .lib (cadence liberate)
- Abstract .lef generation (cadence abstract)



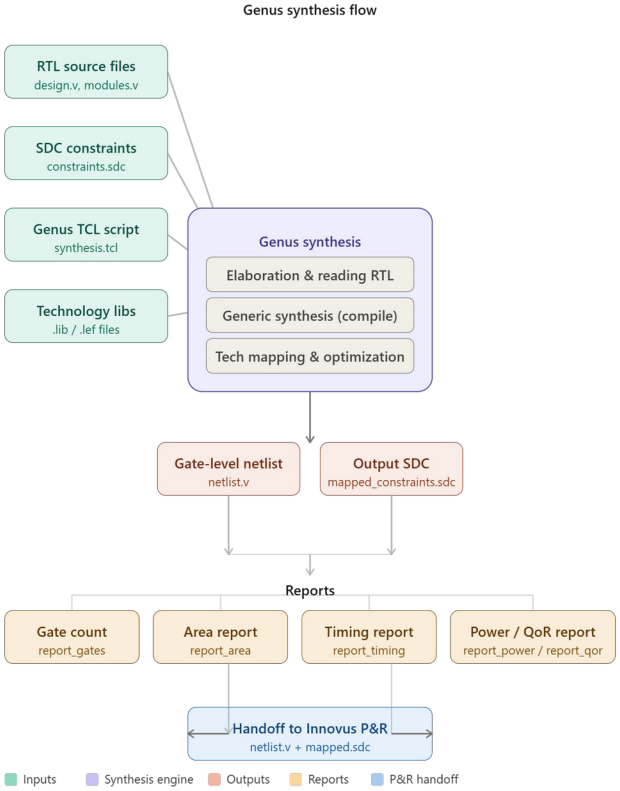
layout of dfxtp (register) standard cell

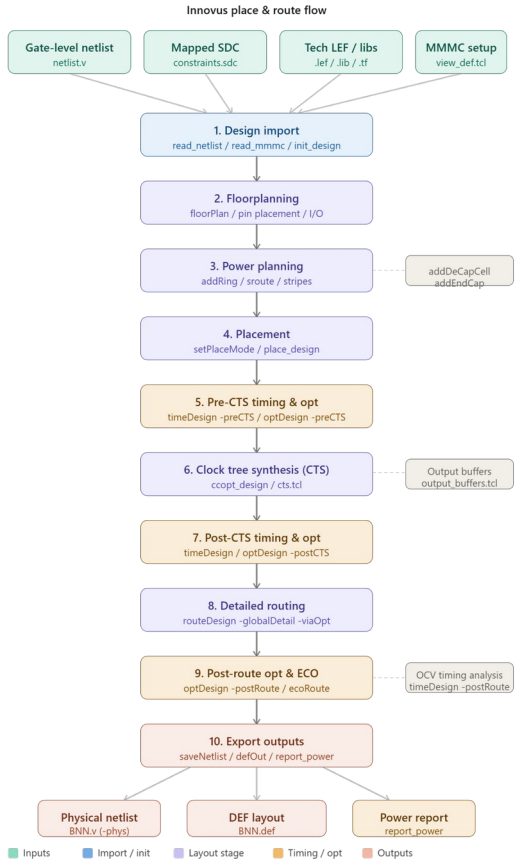


layout of buffer_16 standard cell

iopin
ipin
layout_copy
mem16_dp_AW4_DW4_INIT_CLEAR0
mem16_dp_AW4_DW4_INIT_CLEAR0_409
mem16_dp_AW4_DW16_INIT_CLEAR1
mem_controller_rd_AW4_DW4
mem_controller_rd_AW4_DW4_410
mux2
mux2i
nand2_1
nand2_2
nand2_4
nand2_16
nand2b_1
nand3_1
nand3b_1
nmos2v_CDNS_743650231470
nmos2v_CDNS_743650231476
nmos2v_CDNS_7436502314716
nmos2v_CDNS_7436502314717
nmos2v_CDNS_7436502314720
nmos2v_CDNS_7436502314738
nor2_1
nor2_4
nor2_16
nor3_1
o21a_1
o21ai_1
opin
pmos2v_CDNS_743650231471

Genus and Innovus Flow





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Top Level Layout

